THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a rhombus ABCD, $\angle A = (3x + 15)^\circ$ and $\angle B = (6x + 39)^\circ$. The value of x is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 18

2. In a kite ABCD with AB = AD and CB = CD, $\angle A = 72^\circ$ and $\angle C = 98^\circ$. The angle $\angle B$ is: 1

(a) 190° **(b)** 95° **(c)** 85° **(d)** 105°

3. Three exterior angles of a quadrilateral are 70°, 110° and 85°. The fourth exterior angle is: 1

(a) 105° **(b)** 95° **(c)** 85° **(d)** 120°

4. In which quadrilateral are the diagonals perpendicular but do not bisect each other? 1

(a) parallelogram **(b)** rectangle **(c)** square **(d)** kite

5. The angles of a quadrilateral are x°, $(x + 30)^\circ$, $(x + 60)^\circ$ and $(x + 90)^\circ$. The value of x is: 1

(a) 15 **(b)** 45 **(c)** 75 **(d)** 55

6. The adjacent sides of a parallelogram are in the ratio 4:7 and its perimeter is 66 cm. The longer side is: 1

(a) 42 cm **(b)** 33 cm **(c)** 12 cm **(d)** 21 cm

7. The diagonals of a parallelogram ABCD meet at O. If AO $= (4x + 2)$ cm and OC $= (3x + 9)$ cm, then AC is: 1

(a) 60 cm **(b)** 30 cm **(c)** 62 cm **(d)** 7 cm

8. In a rhombus ABCD, $\angle A = (2x + 40)^\circ$ and $\angle B = (3x + 50)^\circ$. The value of x is: 1

(a) 19 **(b)** 17 **(c)** 18 **(d)** 22

9. In a kite ABCD with AB = AD and CB = CD, $\angle A = 90^\circ$ and $\angle C = 66^\circ$. The angle $\angle B$ is: 1

(a) 92° **(b)** 102° **(c)** 112° **(d)** 204°

10. Three exterior angles of a quadrilateral are 92°, 88° and 105°. The fourth exterior angle is: 1

(a) 85° **(b)** 100° **(c)** 75° **(d)** 65°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The diagonals of a parallelogram ABCD meet at O with AO $= (5x + 2)$ cm and OC $= (4x + 6)$ cm. Find x and AC. 2

12. The exterior angles at three vertices of a quadrilateral are 82°, 76° and 120°. Find the fourth exterior angle and the fourth interior angle. 2

13. The angles of a quadrilateral are in the ratio 3:5:6:10. Find the largest angle. 2

14. The diagonals of a parallelogram ABCD meet at O with AO $= (7x + 1)$ cm and OC $= (5x + 9)$ cm. Find x and AC. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a quadrilateral are $x°$, $(x + 12)°$, $(x + 24)°$ and $(x + 36)°$. Find all the angles. 3

16. The adjacent sides of a parallelogram are in the ratio 5:7 and its perimeter is 72 cm. Find the lengths of its sides. 3

17. In a rectangle ABCD, $\angle ACB = 38°$. Find $\angle CAB$, $\angle AOB$ and $\angle ODC$ where O is the point of intersection of the diagonals. 3

OR

The angles of a quadrilateral are $x°$, $(x + 18)°$, $(x + 36)°$ and $(x + 54)°$. Find all the angles. 3

18. The adjacent sides of a parallelogram are in the ratio 4:9 and its perimeter is 52 cm. Find the lengths of its sides. 3

OR

In a rectangle ABCD, $\angle ACB = 52°$. Find $\angle CAB$, $\angle AOB$ and $\angle ODC$ where O is the point of intersection of the diagonals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (4x + 2)°$ and $\angle OBA = (5x - 2)°$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

OR

The diagonals of a rectangular frame ABCD are AC $= (4x + 6)$ cm and BD $= (2x + 16)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

20. ABCD is a rhombus whose diagonals meet at O. $\angle OAB = (2x + 20)°$ and $\angle OBA = (4x - 8)°$. Find x, $\angle A$ and $\angle B$ of the rhombus. 5

OR

The diagonals of a rectangular frame ABCD are AC $= (6x + 1)$ cm and BD $= (4x + 11)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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